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Roll no. - 69

In [1]:
```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report

# Load the dataset
iris = load_iris()

# Create a clean pandas DataFrame to look at the data
df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
df['species'] = iris.target

# Show the first 5 rows
df.head()
```

Out[1]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

In [2]:
```
# Separate features (X) and target (y)
X = df.drop(columns=['species'])
y = df['species']

# Split: 80% for training, 20% for testing
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, rando

print(f"Training set size: {X_train.shape[0]} samples")
print(f"Testing set size: {X_test.shape[0]} samples")
```

```
Training set size: 120 samples
Testing set size: 30 samples
```

In [3]:
```
# Initialize the scaler
scaler = StandardScaler()

# Fit and transform the training data, and transform the test data
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

In [4]:
```
# Initialize the model
model = KNeighborsClassifier(n_neighbors=3)

# Train the model
```

```
model.fit(X_train_scaled, y_train)
print("Model training complete!")
```

```
Model training complete!
```

In [5]:
```
# Predict on the test set
y_pred = model.predict(X_test_scaled)

# Calculate and print accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy * 100:.2f}%\n")

# Print a detailed classification report
print("Classification Report:")
print(classification_report(y_test, y_pred, target_names=iris.target_names))
```

```
Model Accuracy: 100.00%

Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      1.00      1.00         9
   virginica       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

In [ ]:
```

```